

Team Number 10

Project Title: Intelligent and Collaborative Autonomous Robots for Real-Time Applications

Report Date: 10/21/2024

Part one: Progress status as a team

All team members have detailed tasks listed on Teams Planner: YES

Please answer the following questions:

1. Have you met as a team this past week? If yes, give date/time, and the members attended the meeting. If no, explain.
 - a. Yes team met on 10/22 at 4pm

2. Have you met with the sponsor as a team? If yes, give date/time, and the member attended the meeting. If no, explain.
 - a. Yes team met in the lab: 10/22 at 5:00pm and 10/24 at 6:00 pm

3. Describe verbally the tasks completed the past week and the challenges faced.
 - a. We were able to run the object detection model on the Jetson Nano and recorded the results.
 - b. We had trouble importing and installing the proper libraries so that we could use the Jetson on different nanos of the robot and their cameras.
 - c. We still are trying to optimize the model running on the dog as well as transmit the model over a UDP stream

4. Describe the tasks to be completed the coming week.
 - a. Produce faster object detection results
 - b. Successfully transfer the SDKs and files to the Jetson

Copy the following table from the last week's report (10/14)

	# Tasks completed past week	# Tasks not completed	# Tasks for next week
David Uzan	0	0	2
Ian Simmons	0	2	2
John Sinfuego	0	0	2

Fill in the following table for this week: (10/21)

	# Tasks completed past week	# Tasks not completed	# Tasks for next week
David Uzan	2	0	1
Ian Simmons	1	0	2
John Sinfuego	2	0	2

Part 2: Individual Progress Report (each member complete one form)

Name of the Member: Ian Simmons

Report Date: 10/21/2024

List the following:

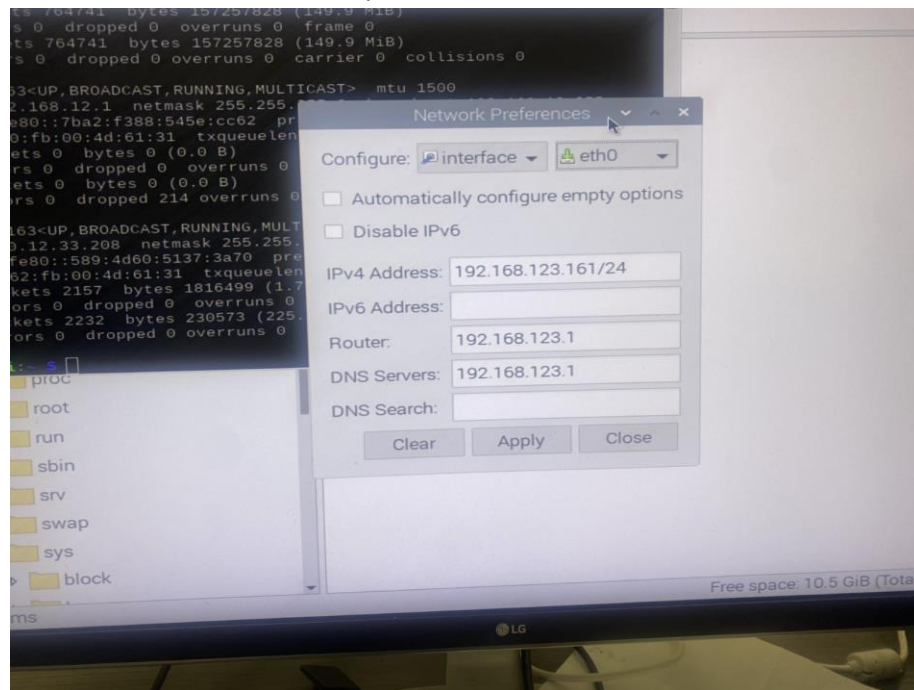
1. All tasks completed the week before the past week (completion date in parenthesis). This is copied from the last report, and if this report is the first one, skip.
2. All tasks completed the past week (completion date in parenthesis)
3. All tasks you are currently working on or planned for this coming week (completion date in parenthesis)

For **each** task you have completed **before the past week**:

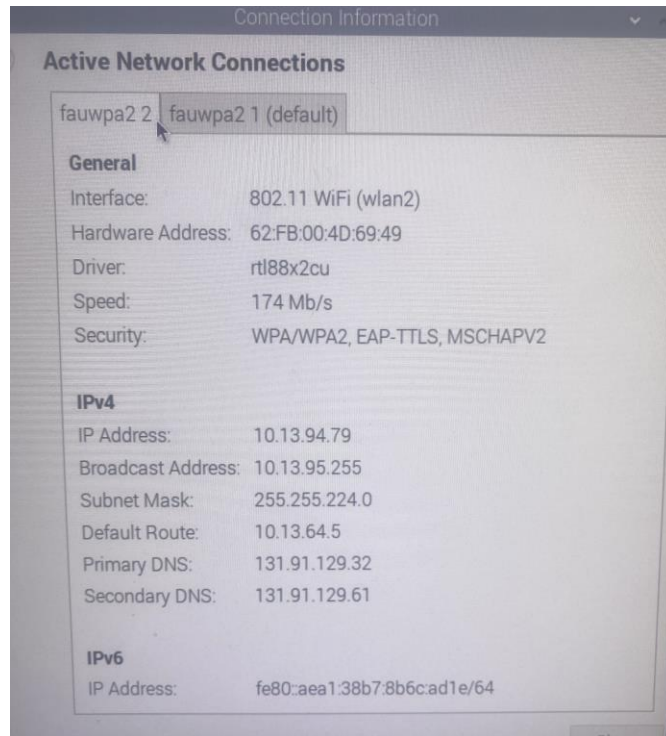
1. Week 8, Task 1 (10/14)
 - a. N/A

For **each** task you have completed **during the past week**:

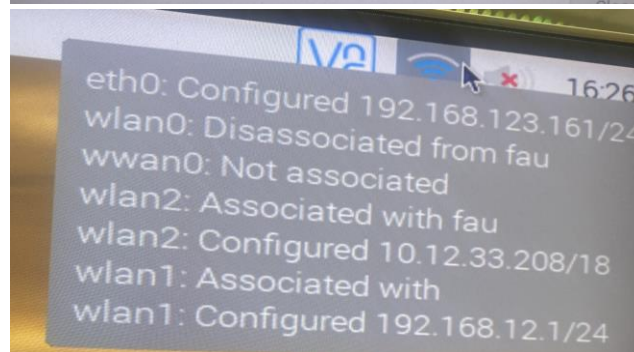
2. Week 9, Task 1 (10/21)
 - a. Over this past week the dog is now able to successfully connect to Wifi. This is done using the Network Manager application on the Raspberry Pi. This will allow us to connect to the school's (or any other) SSID.



b.



c.



- d. The motors are halfway disrupted still and can't operate fully because of the way the robot's internal network operates it may be possible

For **each** task you are currently working on or plan to work on in the next period:

1. **Week 10, Task 1 (10/28)**

- a. The raspberry pi uses a SoC (System on Chip) design so the GPU and CPU are integrated together, currently looking for ways to maximize CPU and GPU usage for our YOLO model

2. **Week 10, Task 2 (10/28)**

- a. Although we can run our model on the head nano, there is no way to do it from a local machine as yet, one of the next steps is to incorporate code in order to utilize the image transmission over UDP. This may also speed up our YOLO model

Part 2: Individual Progress Report (each member complete one form)

Name of the Member: David Uzan

Report Date: 10/21/24

List the following:

1. All tasks completed the week before the past week (completion date in parenthesis). This is copied from the last report, and if this report is the first one, skip.
2. All tasks completed the past week (completion date in parenthesis)
3. All tasks you are currently working on or planned for this coming week (completion date in parenthesis)

For **each** task you have completed **during the before the past week**:

1. Task number/name

N/A

For **each** task you have completed **during the past week**:

1. Task number/name

Personal Week 8 (10-21-24) - Research means to keep motor function with Internet and run model from the robot on a personal computer.

Personal Week 9 (10-21-24) - Run model on Jetson Nano

2. Short description – these descriptions form the basis of the final project report

- a. Implementation, testing, results

As of the last meeting, all I found for a means to connect to the robot over the Internet and keep motor functionality was over a wireless connection. The wireless connection is very unreliable and prone to timing out so wired will have to do for now.

The Jetson Nano was integrated into the robot, replacing the raspberry pi, which yielded great results comparatively. The new runtime was around 700-800 milliseconds, which is close to half of what it was when we ran the model over the pi.

- b. Brief and to point

Wired connection will have to do for now with the robot as a wireless connection is unreliable and poor at staying connected.

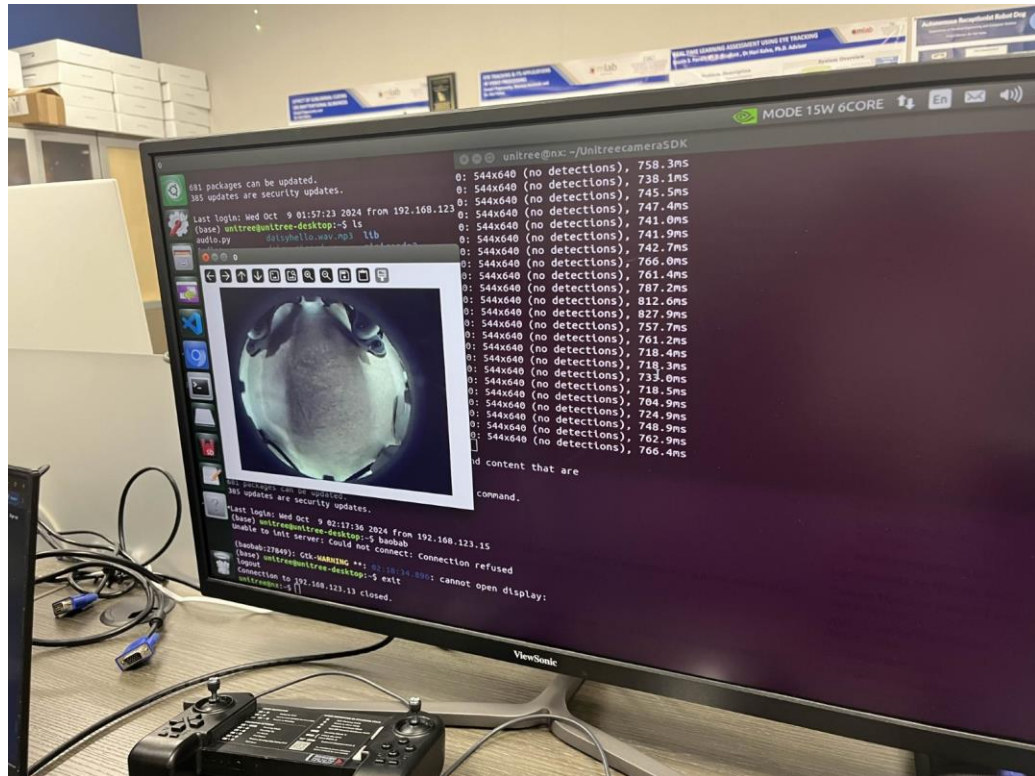
The Jetson Nano drastically improved the runtime of the object detection model compared to previous tests on the raspberry pi.

3. Outcome of the task activity

a. How did the task advance your project?

The runtime of the object detection model now takes less than a second, which is a great reduction but still has some space for improvement.

b. Include screen shots, link to videos, plots, data tables, designs, figures, etc. as evidence of your work on the task



For **each** task you are currently working on or plan to work on in the next period:

1. Task number/name

Personal Week 10 (10-28-24) - Reliably transmit live video from other cameras than front

2. Short description

a. Plans, resources needed, implementation, results

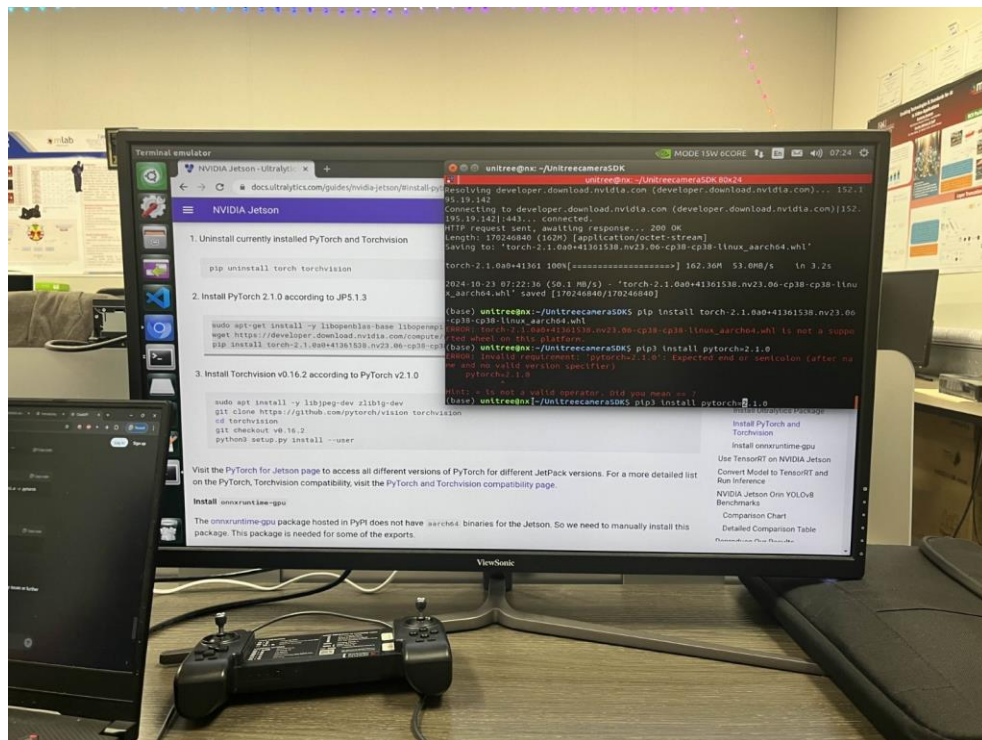
Since different cameras are hosted on different nanos within the robot's system, we must install the supported libraries and programs in everyone that we plan to use. The system requirements necessary to run the Jetson Nano for the cameras are somewhat rigid and we ran into some complications during the installation process.

b. Main challenges, reasons for any delay in completion

Getting the proper versions of libraries and toolkits to help in the running of certain applications on the Jetson Nano is tedious because of the conflicting PyTorch and Python versions.

3. Outcome of the task activity

- a. Include screen shots, link to videos, plots, data tables, designs, figures, etc. as evidence of your work on the task



Part 2: Individual Progress Report (each member complete one form)

Name of the Member: John Angelo Sinfuego

Report Date: 10/21/24

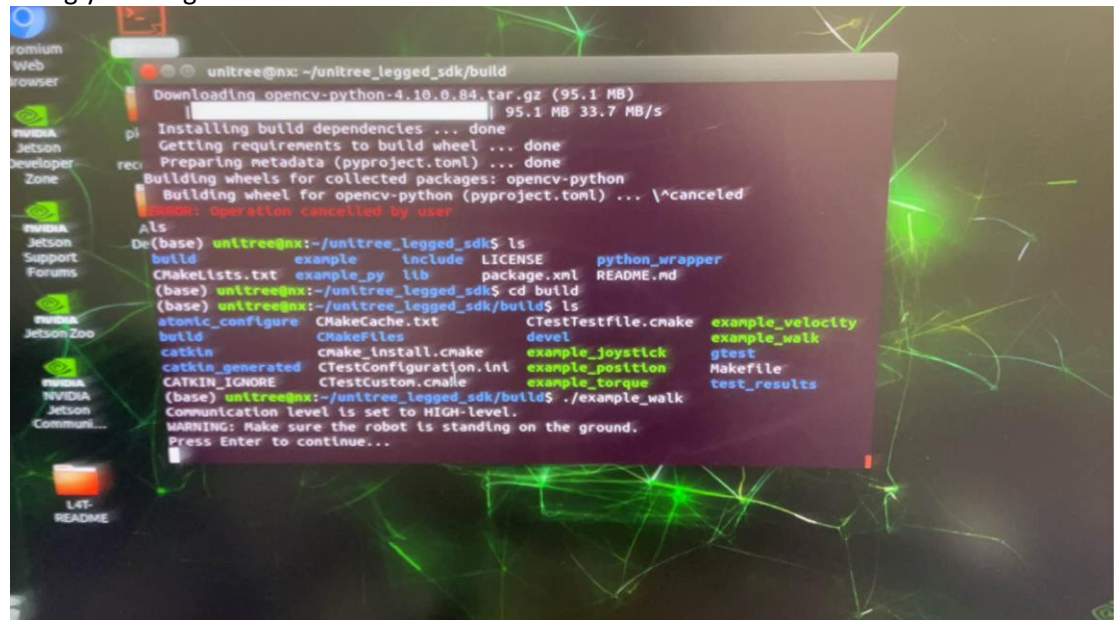
For **each** task you are currently working on or plan to work on in the next period:

1. Task number/name

General Task 8 10-21-24- Optimize the running of yolo

Personal Week 8 10-21-24- 2. Send commands to robot via powershell

b. We are still research methods to optimize the running of yolo through the uni-go1 camera. I however, sent commands to the robot via powershell runnng Unitree_legged_sdk. When the certain commands were sent, the robot would move accordingly to the given code.



3.

Name of the Member: John Angelo Sinfuego

Report Date: 10/14/24

For **each** task you are currently working on or plan to work on in the next period:

2. Task number/name

General Task 6 10-7-24- Incorporate yolo on dog

Personal Week 8 10-7-24- - 2. Research more and possibly start sending command ot dog

a. For our general tasks, my group members and I successfully ran YOLO on uni-go1. We can now transition into our next steps. I also research more on unitree_legged_sdk and will be incorporating that soon as well.

b. N/A solution to make the speed and run time faster.

4. Outcome of the task activity